

Options Strategies and Quant Trading

<https://gemini.google.com/app/25f4dfcd91b65794>

User prompt: what are the various options strategy ? Also what are the various important things quants and systematic traders watchout for in options trading?

Response: Core Options Strategies Options strategies range from simple directional bets to complex, multi-leg structures designed to isolate specific market variables.

- **Directional Plays:** Buying outright Calls (bullish) or Puts (bearish). These provide leveraged upside with capped downside, though they constantly lose value to time decay.
- **Income Generation:** Covered Calls and Cash-Secured Puts. Traders collect upfront premium in exchange for capping their upside or agreeing to buy the stock at a lower predetermined price.
- **Vertical Spreads:** Bull/Bear Call and Put Spreads. By simultaneously buying and selling options at different strikes, traders cap their maximum risk and reward, significantly reducing the upfront cost of the trade.
- **Volatility & Range Strategies:** Straddles and Strangles profit from explosive price moves in *either* direction (long volatility). Conversely, Iron Condors and Butterflies are short-volatility trades that profit when the underlying asset remains stagnant within a tight range.
- **Advanced Quantitative Strategies:** Dispersion trading (exploiting the implied correlation between an index and its individual components), calendar spreads (exploiting differences in time decay across expirations), and volatility surface anomaly arbitrage.

What Quants & Systematic Traders Monitor Quantitative and systematic traders typically avoid directional guessing. Instead, they engineer mathematical models to isolate statistical anomalies, manage automated risk, and trade volatility as an asset class itself.

- **The Volatility Surface & Skew:** Implied volatility (IV) is not a single number; it varies by strike price (creating a "volatility smile" or "skew") and expiration date (term structure). Quants mathematically map this 2D surface to find mispriced options relative to historical probabilities.
- **Implied vs. Realized Volatility:** The market generally overestimates future risk. A foundational quant edge involves capturing the "Volatility Risk Premium" by exploiting the spread between Implied Volatility (market expectation) and Realized Volatility (actual historical movement).
- **Dynamic Hedging via Greeks:** Systematic traders often run "delta-neutral" portfolios, relying on the Greeks to isolate specific risks:
 - **Delta:** Directional risk. Quants continuously buy or sell the underlying asset to keep the portfolio's overall Delta at zero.
 - **Gamma:** The rate of change of Delta. High Gamma means the Delta fluctuates rapidly, requiring the algorithmic system to re-hedge constantly.
 - **Theta:** The mathematical daily cost of holding the option (time decay).
 - **Vega:** Exposure to changes in implied volatility. "Long vol" funds trade specifically to maximize Vega.

- **Hedging Costs & Execution Slippage:** Because delta-neutral strategies require constantly trading the underlying stock to stay balanced, the bid-ask spread can rapidly eat away profits. Quants write execution algorithms to optimize the exact frequency of re-hedging to minimize trading friction.
- **Model Risk & Fat Tails:** Standard pricing models (like Black-Scholes) assume asset prices follow a lognormal distribution. In reality, financial markets exhibit "fat tails," meaning extreme outlier events happen more frequently than a bell curve predicts. Systematic traders adapt their algorithms using advanced models (like SABR or GARCH) to price in tail risks and prevent catastrophic drawdowns during black swan events.

Sources

1. [Options Greeks Explained: Delta, Gamma, Theta - Quantt](#)
 2. [Systematic risk management](#)
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User prompt: do indian brokers apart from OHLCV provide any of the greeks and IV etc in websocket or do they provide any detail which can be help in dynamically compute those values locally?

Response: Indian Brokers Offering Pre-Calculated Greeks & IV

Most major Indian brokers rely on users calculating their own Greeks, but a few have started providing them directly through their API infrastructure:

- **Upstox:** Natively streams pre-calculated Greeks and Implied Volatility (IV) directly through their Live Market Data WebSocket feed. The data payload includes specific optionGreeks objects containing iv, delta, theta, gamma, and vega alongside the standard tick data.
- **Dhan:** Recently launched a dedicated "Option Chain with Greeks" REST API endpoint. While not a push-based WebSocket feed, this single API call returns the entire option chain with real-time pre-calculated Greeks, IV, Open Interest (OI), and bid/ask depths, eliminating the need for local computation.
- **Zerodha / Fyers / Angel One:** Primarily stream raw market data (LTP, Best Bid/Ask, OHLCV, and OI) over their WebSockets. Traders must build a local pricing engine to derive Greeks.

Dynamic Data Provided for Local Computation

To compute Greeks and IV locally using a pricing model (like Black-Scholes or Black-76), systematic traders rely on the standard WebSocket Level 1 or Level 2 (Market Depth) tick feeds to capture the dynamic variables:

- **Live Option Price:** The Last Traded Price (LTP) or the mid-point of the Best Bid/Ask spread, extracted directly from the option's WebSocket tick.
- **Live Underlying Price:** You must actively subscribe to the underlying asset (e.g., Nifty Spot index or the near-month Future) in the same WebSocket connection. Options pricing models require the real-time underlying price to function accurately.
- **Timestamp:** The exchange timestamp provided in the tick data is used to calculate the exact Time to Expiry (often down to the minute or second) for intraday precision.

The Local Computation Process

You combine the dynamic WebSocket data with static variables tracked locally on your server:

1. **Static Variables:** You maintain the Strike Price (from the broker's instrument master file) and the Risk-Free Interest Rate (typically the RBI 91-day T-bill yield, currently around 6.5% - 7%).
2. **Deriving Implied Volatility (IV):** Because the Black-Scholes formula solves for the option price based on volatility, you must reverse-engineer it. You feed the Live Option Price, Live Underlying Price, Strike, Time to Expiry, and Risk-Free Rate into a root-finding mathematical algorithm (like the Newton-Raphson method) to extract the IV.
3. **Calculating the Greeks:** Once your algorithm calculates the IV, you immediately plug that IV value back into the standard derivative formulas to dynamically generate your real-time Delta, Gamma, Theta, and Vega.

Sources

1. [Market Data Feed - Developer API - Upstox](#)
 2. [Understanding Option Chain with Greeks on DhanHQ API](#)
 3. [How to view & read option chain on Kite? - Support Zerodha](#)
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User prompt: how about fyers?

Response: When it comes to **Fyers**, they actually occupy a highly useful middle ground between providing everything dynamically and requiring you to build everything from scratch.

With the release of their **Fyers API v3**, they do provide pre-calculated Greeks and Implied Volatility, but **not** directly through the WebSocket stream.

Here is exactly how Fyers handles options data for systematic traders:

1. Pre-Calculated Greeks via REST API (Pull Data)

Fyers provides a dedicated **GetOptionChain REST API endpoint**. When you make a request to this endpoint for a specific underlying asset (like NIFTY), it returns the entire option chain snapshot.

- **What it includes:** Alongside the standard pricing and Open Interest (OI) data, this payload natively includes pre-calculated **Delta, Gamma, Theta, Vega, and Implied Volatility (IV)** for every strike.
- **The Catch:** Because it is a REST API, you have to actively "poll" or request it. You cannot ping a REST API 100 times a second without hitting rate limits, so this is best used for periodic snapshots rather than microsecond-level algorithmic execution.

2. WebSocket Feed (Push Data)

The Fyers WebSocket (Data API) is designed for raw speed and low latency.

- **What it streams:** It streams standard Level 1 (LTP, Volume, OI) and Level 2 (up to 50 levels of Bid/Ask Market Depth) data.

- **What it lacks:** It **does not** stream the pre-calculated Greeks or IV dynamically on every tick.

How Quants Trade on Fyers (The Hybrid Approach)

Because Fyers provides both a robust REST API and a fast WebSocket, systematic traders usually build a hybrid system to get the best of both worlds:

1. **Baseline Synchronization:** The algorithm polls the `getOptionChain` REST API every few seconds or minutes to get a baseline "truth" for the Implied Volatility and Greeks across the entire chain.
2. **Live WebSocket Subscriptions:** Simultaneously, the algorithm subscribes to the Fyers WebSocket for the specific Option strikes being traded, *plus* the underlying index (e.g., Nifty Spot).
3. **Local Dynamic Recalculation:** Between the REST API polling intervals, the algorithm uses the millisecond-level LTP and Bid/Ask updates from the WebSocket to constantly recalculate the Greeks locally. Because the system already has a recent IV baseline from the REST API, the local Black-Scholes math requires far less computational power to update the Delta and Gamma on the fly.

This makes Fyers a very popular choice for option sellers and systematic traders in India, as it provides the raw materials for low-latency local computation while still offering an authoritative Option Chain API to keep your local mathematical models calibrated.

Sources

1. [OptionX vs Fyers Scalper Terminal: Why Active Options](#)

User prompt: I plan to build an autonomous system. That is self improving in nature. It has lets say 4-5 strategies you mentioned above. Based on various tech factors , global markets etc i want to build a system that actively trades without my intervention. Service 1/Script1 - Pulls live data via websocket and periodically pull greeks OI etc via Rest API Service 2 - Periodically pulls news and other non tech data that the system needs. Service 3 does the local computation as required Agent1 - Decides positional trade for Nifty/Stocks/ Banknifty etc to decide which strategy is best fit and selects appropriate strike prices or Future contracts to deploy the required strategy.. No preference if positional or intraday as long as the fitment is best. The max loss and max win is pre-defined. Agent2/Script -- Periodically checks for super confident setups to enter naked option buys .. this shouldnt impact the trades taken by agent 1. This should be based on news/OI, greeks, or whatever. i Have no clue yet. This has lets say 10% of the overall portfolio and doesnt have to be positional Agent 3 - Which periodically adjusts the SL and trails and pre-exit if the trade becomes super weak with backed data only. Agent 4 identifies the best candidate or decides the universe to focus within a week. Health scripts with an ai agent -- The script runs checks across system for failures or misfunction and then invokes the agent to fix the problem if the agent is not able to fix it triggers the alarm to me via telegram/pagerduty or someway The systems records all the reasoning somewhere. Another agent analyzes what the agent did well or missed to do and records the area of improvements.. For something fatal it triggers alarm.. This is executed close of trading day. I dont want this to be purely script based or purely AI based.. We can tweak the same as well. Asks from you : 1. Help me design a system 2. Help with which strategies basket are good fit 3. Should have necessary adaptations for smoother markets of MCX as well gap and manipulated derivatives of indian stock exchanges of NSE 4. This will be deployed in AWS and ORacle to leverage free lambda limits and Oracle VPS free tier for PAYG .. but i dont mind having some overall cost 5. Position sizing. 6. capital

allocation for 1-2 strategies that agent 1 may select and some 10% naked option buys. 7. The system should beat the index and not ignore high conviction parabolic moves but majorly enters predefined loss and profit strategies I need to have a solid workflow. A system

Response: I've put together a research plan. If you need to update it in any way, let me know.

User prompt: Start research

Response: I've completed your research. Feel free to ask me follow-up questions or request changes.

User prompt: Enhance the research with within what should be agent and what should be predefined scripts. I just don't want agents on flow which requires extreme fast decision else I prefer agents. We have to design it accordingly. If there is an AI agent then the script should act as fallback. If there is script and AI agent should act as it fallback. This needs to be super duper careful. Also apart from designing for CAS. Why not take mini trades to take advantage of CAS this could be agent2. We can close agent2 trades if required to avoid CAS related impact. Take a small naked option buy and let the money spike or go to loss more like a lottery. A 50% win rate with data backed approach will be awesome. Also think of factors I have not thought about that are getting implemented in latest tech advancements in 2026 by folks with good tech skills which doesn't necessarily require a high compute cost.

Response: Sure, I have updated the system design to clearly delineate between deterministic scripts and asynchronous AI agents, implemented strict fallback mechanisms for both, added the 2026 low-compute tech stack (DuckDB and Polars), and integrated the data-backed CAS "lottery" strategy.

Here is your updated research:

I've reorganized the architecture to clearly separate the deterministic scripts from the AI agents, ensuring that time-critical processes are handled instantly while AI focuses on heavy evaluation. I added the bidirectional fallback mechanism to prevent system halts. I also updated the technology stack to include DuckDB and Polars for efficient, low-compute processing, and inserted the detailed strategy for taking advantage of CAS imbalances using your 10% naked buy allocation. Let me know if you need anything else modified.
